

# Imprinting (psychology)

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In psychology and ethology, **imprinting** is any kind of phase-sensitive learning (learning occurring at a particular age or a particular life stage) that is rapid and apparently independent of the consequences of behaviour. It was first used to describe situations in which an animal or person learns the characteristics of some stimulus, which is therefore said to be "imprinted" onto the subject. Imprinting is hypothesized to have a critical period.

## Contents

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**Filial imprinting**

**Sexual imprinting**

**Limbic imprinting**

**Westermarck effect**

**Baby duck syndrome**

**See also**

**References**

**Further reading**

**External links**

## Filial imprinting

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The best-known form of imprinting is *filial imprinting*, in which a young animal narrows its social preferences to an object (typically a parent) as a result of exposure to that object. It is most obvious in nidifugous birds, which imprint on their parents and then follow them around.<sup>[1]</sup> It was first reported in domestic chickens, by Sir Thomas More in 1516 as described in his treatise *Utopia*, 350 years earlier than by the 19th-century amateur biologist Douglas Spalding. It was rediscovered by the early ethologist Oskar Heinroth, and studied extensively and popularized by his disciple Konrad Lorenz working with greylag geese.<sup>[2]</sup>

Lorenz demonstrated how incubator-hatched geese would imprint on the first suitable moving stimulus they saw within what he called a "critical period" between 13 and 16 hours shortly after hatching. For example, the goslings would imprint on Lorenz himself (to be more specific, on his wading boots), and he is often depicted being followed by a gaggle of geese who had imprinted on him. Lorenz also found that the geese could imprint on inanimate objects. In one notable experiment, they followed a box placed on a model train in circles around the track.<sup>[2]</sup> Filial imprinting is not restricted to non-human animals that are able to follow their parents, however.

The filial imprinting of birds was a primary technique used to create the movie *Winged Migration* (*Le Peuple Migrateur*), which contains a great deal of footage of migratory birds in flight. The birds imprinted on handlers, who wore yellow jackets and honked horns constantly. The birds were then trained to fly along with a variety of aircraft, primarily ultralights.

The Italian hang-glider pilot Angelo d'Arrigo extended this technique. D'Arrigo noted that the flight of a non-motorised hang-glider is very similar to the flight patterns of migratory birds; both use updrafts of hot air (thermal currents) to gain altitude that then permits soaring flight over distance. He used this to reintroduce threatened species of raptors.<sup>[3]</sup> Because birds hatched in captivity have no mentor birds to teach them traditional migratory routes, D'Arrigo hatched chicks under the wing of his glider and they imprinted on him. Then, he taught the fledglings to fly and to hunt. The young birds followed him not only on the ground (as with Lorenz) but also in the air as he took the path of various migratory routes. He flew across the Sahara and over the Mediterranean Sea to Sicily with eagles, from Siberia to Iran (5,500 km) with a flock of Siberian cranes, and over Mount Everest with Nepalese eagles. In 2006, he worked with a condor in South America.<sup>[3]</sup>



Imprinted Canada geese (*Branta canadensis*) and common crane (*Grus grus*) flying with an ultralight aircraft

In a similar project, orphaned Canada geese were trained to their normal migration route by the Canadian ultralight enthusiast Bill Lishman, as shown in the fact-based movie drama *Fly Away Home*.

Chicks of domestic chickens prefer to be near large groups of objects that they have imprinted on. This behaviour was used to determine that very young chicks of a few days old have rudimentary counting skills. In a series of experiments, they were made to imprint on plastic balls and could figure out which of two groups of balls hidden behind screens had the most balls.<sup>[4]</sup>

American coot mothers have the ability to recognize their chicks by imprinting on cues from the first chick that hatches. This allows mothers to distinguish their chicks from parasitic chicks.

The peregrine falcon has also been known to imprint on specific structures for their breeding grounds such as cliff sides and bridges and thus will favour that location for breeding.<sup>[5]</sup>

## **Sexual imprinting**

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**Sexual imprinting** is the process by which a young animal learns the characteristics of a desirable mate. For example, male zebra finches appear to prefer mates with the appearance of the female bird that rears them, rather than that of the birth parent when they are different.<sup>[6]</sup>

Sexual attraction to humans can develop in non-human mammals or birds as a result of sexual imprinting when reared from young by humans. One example is London Zoo female giant panda Chi Chi. When taken to Moscow Zoo for mating with the male giant panda An An, she refused his attempts to mate with her, but made a full sexual self-presentation to a zookeeper.<sup>[7][8]</sup>

It commonly occurs in falconry birds reared from hatching by humans. Such birds are called "imprints" in falconry. When an imprint must be bred from, the breeder lets the male bird copulate with their head while they are wearing a special hat with pockets on to catch the male bird's semen. The breeder then courts a suitable imprint female bird (including offering food, if it is part of that species's normal courtship). At "copulation", the breeder puts the flat of one hand on the female bird's back to represent the weight of a male bird, and with the other hand uses a pipette, or a hypodermic syringe without a needle, to squirt the semen into the female's cloaca.<sup>[9][10]</sup>

Sexual imprinting on inanimate objects is a popular theory concerning the development of sexual fetishism.<sup>[11]</sup> For example, according to this theory, imprinting on shoes or boots (as with Konrad Lorenz's geese) would be the cause of shoe fetishism.

## Limbic imprinting

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Some suggest that prenatal, perinatal and post-natal experiences leave imprints upon the limbic system, causing lifelong effects and this process is identified as limbic imprinting.<sup>[12]</sup> The term is also described as the human emotional map, deep-seated beliefs, and values that are stored in the brain's limbic system and govern people's lives at the subconscious level.<sup>[13]</sup> It is one of the suggested explanations for the claim that the experiences of an infant, particularly during the first two years of his life, contribute to his lifelong psychological development.<sup>[14]</sup> Imprinted genes can have astounding effects on body size, brain size, and the process in which the brain organizes its processes. Evolutionary trends within the animal kingdom have been shown to show substantive increase in the fore-brain particularly towards the limbic system, this evolution has even been thought of to have a mutative effect on the brain size trickling down the human ancestry.<sup>[15]</sup>

## Westermarck effect

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*Reverse* sexual imprinting is also seen in instances where two people who live in domestic proximity during the first few years in the life of either one become desensitized to later close sexual attraction. This phenomenon, known as the Westermarck effect, was first formally described by Finnish anthropologist Edvard Westermarck in his book *The History of Human Marriage* (1891). The Westermarck effect has since been observed in many places and cultures, including in the Israeli kibbutz system, and the Chinese shim-pua marriage customs, as well as in biological-related families.

In the case of the Israeli kibbutzim (collective farms), children were reared somewhat communally in peer groups, based on age, not biological relation. A study of the marriage patterns of these children later in life revealed that out of the nearly 3,000 marriages that occurred across the kibbutz system, only fourteen were between children from the same peer group. Of those fourteen, none had been reared together during the first six years of life. This result provides evidence not only that the Westermarck effect is demonstrable but that it operates during the period from birth to the age of six.<sup>[16]</sup> However, Eran Shor and Dalit Simchai claimed that the case of the kibbutzim actually provides little support for the Westermarck effect.<sup>[17]</sup>

When proximity during this critical period does not occur—for example, where a brother and sister are brought up separately, never meeting one another—they may find one another highly sexually attractive when they meet as adults.<sup>[18]</sup> This phenomenon is known as genetic sexual attraction. This observation supports the hypothesis that the Westermarck effect evolved because it suppressed inbreeding. This attraction may also be seen with cousin couples.

Sigmund Freud argued that as children, members of the same family naturally lust for one another, making it necessary for societies to create incest taboos,<sup>[19]</sup> but Westermarck argued the reverse, that the taboos themselves arise naturally as products of innate attitudes. Steven Pinker has written that Freud's conception of an urge to incest may have derived from Freud's own erotic reaction to his mother as a boy (attested in Freud's own writings), and speculates that Freud's reaction may have been due to lack of intimacy with his mother in early childhood, as Freud was wet-nursed.<sup>[20]</sup>

## Baby duck syndrome

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In human–computer interaction, **baby duck syndrome** denotes the tendency for computer users to "imprint" on the first system they learn, then judge other systems by their similarity to that first system. The result is that "users generally prefer systems similar to those they learned on and dislike unfamiliar systems".<sup>[21]</sup> The issue may present itself relatively early in a computer user's experience, and it has been observed to impede education of students in new software systems or user interfaces.<sup>[22]</sup>



Ducklings following their mother.

## See also

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- Ivan Pavlov
- Kin recognition
- Kin selection
- Attachment theory
- Imprinting (organizational theory)
- Psychical inertia

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